Prepare Sample Data To Practice SQL Skill.

Sample Table – Worker

WORKER_ID	FIRST_NAME	LAST_NAME	SALARY	JOINING_DATE	DEPARTMENT
001	Monika	Arora	100000	2014-02-20 09:00:00	HR
002	Niharika	Verma	80000	2014-06-11 09:00:00	Admin
003	Vishal	Singhal	300000	2014-02-20 09:00:00	HR
004	Amitabh	Singh	500000	2014-02-20 09:00:00	Admin
005	Vivek	Bhati	500000	2014-06-11 09:00:00	Admin
006	Vipul	Diwan	200000	2014-06-11 09:00:00	Account
007	Satish	Kumar	75000	2014-01-20 09:00:00	Account
008	Geetika	Chauhan	90000	2014-04-11 09:00:00	Admin

Sample Table – Bonus

WORKER_REF_ID	BONUS_DATE	BONUS_AMOUNT
1	2016-02-20 00:00:00	5000
2	2016-06-11 00:00:00	3000
3	2016-02-20 00:00:00	4000
1	2016-02-20 00:00:00	4500
2	2016-06-11 00:00:00	3500

Sample Table – Title

WORKER_REF_ID	WORKER_TITLE	AFFECTED_FROM
1	Manager	2016-02-20 00:00:00
2	Executive	2016-06-11 00:00:00
8	Executive	2016-06-11 00:00:00
5	Manager	2016-06-11 00:00:00
4	Asst. Manager	2016-06-11 00:00:00
7	Executive	2016-06-11 00:00:00
6	Lead	2016-06-11 00:00:00
3	Lead	2016-06-11 00:00:00

Query to create these three tables.

```
CREATE DATABASE workshop;
USE workshop;
```

For worker table:

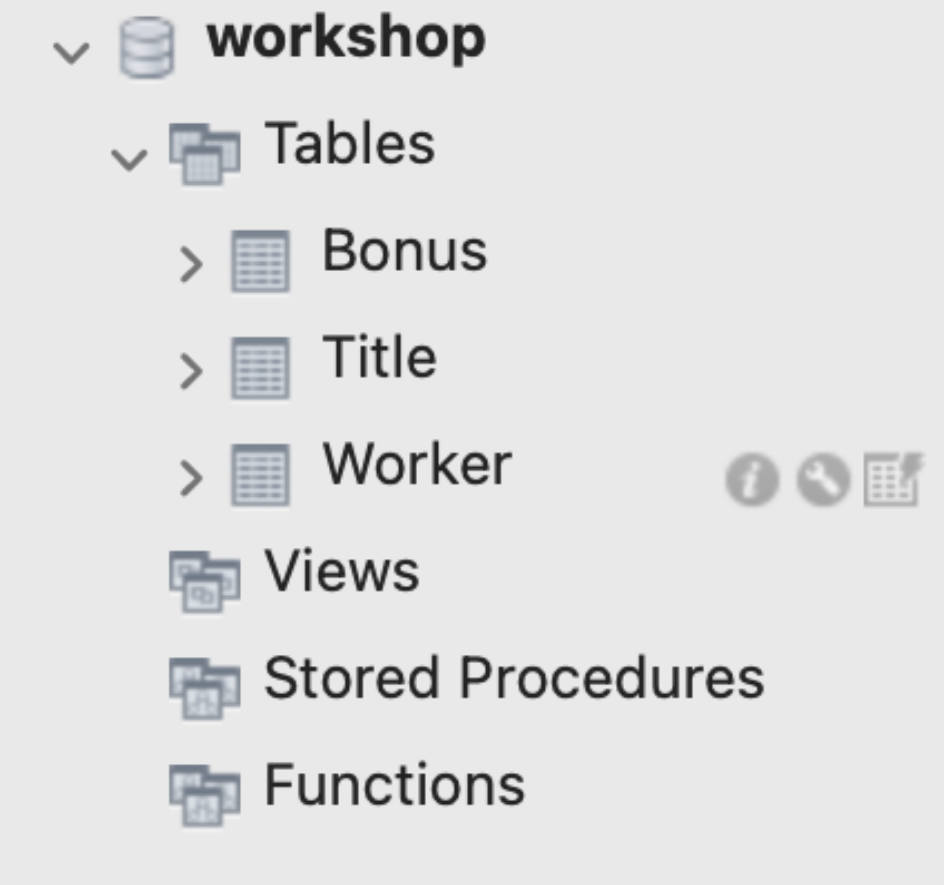
```
CREATE TABLE Worker (
   WORKER_ID INT PRIMARY KEY,
   FIRST_NAME VARCHAR(50),
   LAST_NAME VARCHAR(50),
   SALARY INT,
   JOINING_DATE DATE,
   DEPARTMENT VARCHAR(50)
);
INSERT INTO Worker VALUES
(1, 'Monika', 'Arora', 100000, '2014-02-20', 'HR'),
(2, 'Niharika', 'Verma', 80000, '2014-06-11', 'Admin'),
(3, 'Vishal', 'Singhal', 300000, '2014-02-20', 'HR'),
(4, 'Amitabh', 'Singh', 500000, '2014-02-20', 'Admin'),
(5, 'Vipul', 'Diwan', 200000, '2014-06-11', 'Account'),
(6, 'Satish', 'Kumar', 75000, '2014-01-20', 'Account');
```

For bonus table:

```
CREATE TABLE Bonus (
   WORKER_REF_ID INT,
   BONUS_AMOUNT INT,
   BONUS_DATE DATE,
   FOREIGN KEY (WORKER_REF_ID) REFERENCES Worker(WORKER_ID)
);
INSERT INTO Bonus VALUES
(1, 5000, '2016-02-20'), (2, 3000, '2016-06-11'), (3, 4000, '2016-02-20'),
(1, 4500, '2016-02-20'), (2, 3500, '2016-06-11');
```

For Title table:

```
CREATE TABLE Title (
   WORKER_REF_ID INT,
   WORKER_TITLE VARCHAR(50),
   AFFECTED_FROM DATE,
   FOREIGN KEY (WORKER_REF_ID) REFERENCES Worker(WORKER_ID)
);
INSERT INTO Title VALUES
(1, 'Manager', '2016-02-20'),
(2, 'Executive', '2016-06-11'),
(3, 'Lead', '2016-02-20'),
(4, 'Assistant', '2016-02-20'),
(5, 'Manager', '2016-06-11'),
(6, 'Executive', '2016-01-20');
```



SQL Query Questions for Practice

1. **Write an SQL query to fetch "FIRST_NAME" from Worker table using the alias name as <WORKER_NAME>**

SELECT FIRST_NAME AS WORKER_NAME
FROM Worker;

```
SELECT FIRST_NAME AS WORKER_NAME
FROM Worker;
```

100% 13:2

Result Grid Filter Rows: Search Export:

WORKER_NAME
Monika
Niharika
Vishal
Amitabh
Vipul
Satish

2. **Write an SQL query to fetch "FIRST_NAME" from Worker table in upper case**

SELECT UPPER(FIRST_NAME)
FROM Worker;

```
SELECT UPPER(FIRST_NAME)
FROM Worker;
```

100% 13:2

Result Grid Filter Rows: Search Export:

UPPER(FIRST_NAME)
MONIKA
NIHARIKA
VISHAL
AMITABH
VIPUL
SATISH

3. **Write an SQL query to fetch unique values of DEPARTMENT from Worker table**

SELECT DISTINCT DEPARTMENT
FROM Worker;

```
SELECT DISTINCT DEPARTMENT
FROM Worker;
```

100% 13:2

Result Grid Filter Rows: Search Export:

DEPARTMENT
HR
Admin
Account

4. **Write an SQL query to print the FIRST_NAME and LAST_NAME from the Worker table into a single column COMPLETE_NAME. A space char should separate them**

SELECT CONCAT(FIRST_NAME, ' ', LAST_NAME) AS COMPLETE_NAME
FROM Worker;

```
SELECT CONCAT(FIRST_NAME, ' ', LAST_NAME) AS COMPLETE_NAME
FROM Worker;
```

100% 13:2

Result Grid Filter Rows: Search Export:

COMPLETE_NAME
Monika Arora
Niharika Verma
Vishal Singhal
Amitabh Singh
Vipul Diwan
Satish Kumar

5. Write an SQL query to print all Worker details from the Worker table order by FIRST_NAME Ascending

SELECT *
FROM Worker
ORDER BY FIRST_NAME ASC;

```
SELECT *
FROM Worker
ORDER BY FIRST_NAME ASC;
```

100% 25:3

Result Grid Filter Rows: Search Edit: Export/Import:

WORKER_ID	FIRST_NAME	LAST_NAME	SALARY	JOINING_DATE	DEPARTMENT
4	Amitabh	Singh	500000	2014-02-20	Admin
1	Monika	Arora	100000	2014-02-20	HR
2	Niharika	Verma	80000	2014-06-11	Admin
6	Satish	Kumar	75000	2014-01-20	Account
5	Vipul	Diwan	200000	2014-06-11	Account
3	Vishal	Singhal	300000	2014-02-20	HR
NULL	NULL	NULL	NULL	NULL	NULL

Result Grid
Form Editor

6. Write an SQL query to print all Worker details from the Worker table order by FIRST_NAME Ascending and DEPARTMENT Descending.

SELECT *
FROM Worker
ORDER BY FIRST_NAME ASC, DEPARTMENT DESC;

```
SELECT *
FROM Worker
ORDER BY FIRST_NAME ASC, DEPARTMENT DESC;
```

100% 41:3

Result Grid Filter Rows: Search Edit: Export/Import:

WORKER_ID	FIRST_NAME	LAST_NAME	SALARY	JOINING_DATE	DEPARTMENT
4	Amitabh	Singh	500000	2014-02-20	Admin
1	Monika	Arora	100000	2014-02-20	HR
2	Niharika	Verma	80000	2014-06-11	Admin
6	Satish	Kumar	75000	2014-01-20	Account
5	Vipul	Diwan	200000	2014-06-11	Account
3	Vishal	Singhal	300000	2014-02-20	HR
NULL	NULL	NULL	NULL	NULL	NULL

Result Grid
Form Editor

7. Write an SQL query to print details for Workers with the first name as “Vipul” and “Satish” from Worker table

SELECT *
FROM Worker
WHERE FIRST_NAME IN ('Vipul', 'Satish');

```
SELECT *
FROM Worker
WHERE FIRST_NAME IN ('Vipul', 'Satish');
```

100% 41:3

Result Grid Filter Rows: Search Edit: Export/Import:

WORKER_ID	FIRST_NAME	LAST_NAME	SALARY	JOINING_DATE	DEPARTMENT
5	Vipul	Diwan	200000	2014-06-11	Account
6	Satish	Kumar	75000	2014-01-20	Account
NULL	NULL	NULL	NULL	NULL	NULL

Result Grid Form Editor

8. Write an SQL query to print details of workers excluding first names, “Vipul” and “Satish” from Worker table

SELECT *
FROM Worker
WHERE FIRST_NAME NOT IN ('Vipul', 'Satish');

```
SELECT *
FROM Worker
WHERE FIRST_NAME NOT IN ('Vipul', 'Satish');
```

100% 45:3

Result Grid Filter Rows: Search Edit: Export/Import:

WORKER_ID	FIRST_NAME	LAST_NAME	SALARY	JOINING_DATE	DEPARTMENT
1	Monika	Arora	100000	2014-02-20	HR
2	Niharika	Verma	80000	2014-06-11	Admin
3	Vishal	Singhal	300000	2014-02-20	HR
4	Amitabh	Singh	500000	2014-02-20	Admin
NULL	NULL	NULL	NULL	NULL	NULL

Result Grid Form Editor

9. Write an SQL query to print details of the Workers whose FIRST_NAME contains ‘a’

SELECT *
FROM Worker
WHERE FIRST_NAME LIKE '%a%';

```
SELECT *
FROM Worker
WHERE FIRST_NAME LIKE '%a%';
```

100% 29:3

Result Grid Filter Rows: Search Edit: Export/Import:

WORKER_ID	FIRST_NAME	LAST_NAME	SALARY	JOINING_DATE	DEPARTMENT
1	Monika	Arora	100000	2014-02-20	HR
2	Niharika	Verma	80000	2014-06-11	Admin
3	Vishal	Singhal	300000	2014-02-20	HR
4	Amitabh	Singh	500000	2014-02-20	Admin
6	Satish	Kumar	75000	2014-01-20	Account
NULL	NULL	NULL	NULL	NULL	NULL

Result Grid Form Editor

10. Write an SQL query to print details of the Workers whose SALARY lies between 100000 and 500000

SELECT *
FROM Worker
WHERE SALARY BETWEEN 100000 AND 500000;

```
SELECT *
FROM Worker
WHERE SALARY BETWEEN 100000 AND 500000;
```

100% 40:3

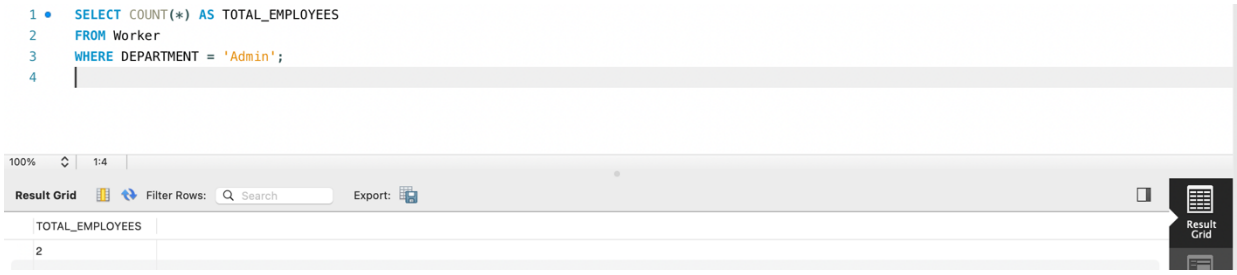
Result Grid Filter Rows: Search Edit: Export/Import:

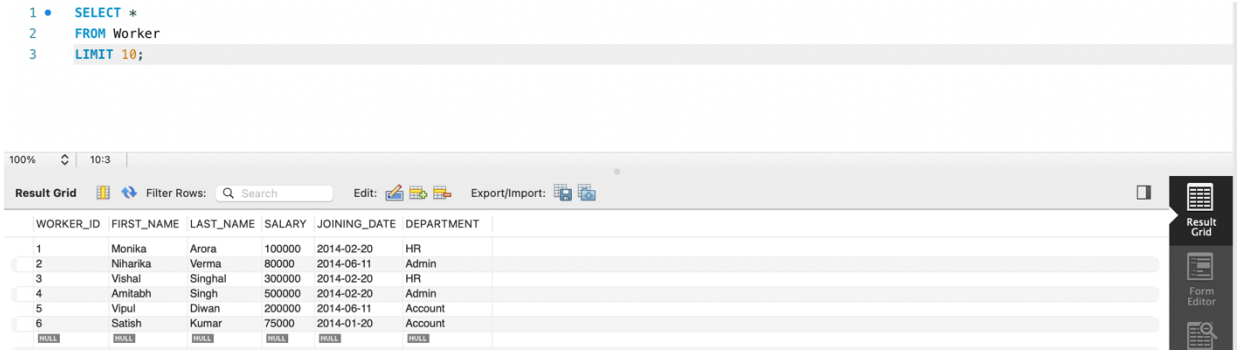
WORKER_ID	FIRST_NAME	LAST_NAME	SALARY	JOINING_DATE	DEPARTMENT
1	Monika	Arora	100000	2014-02-20	HR
3	Vishal	Singhal	300000	2014-02-20	HR
4	Amitabh	Singh	500000	2014-02-20	Admin
5	Vipul	Diwan	200000	2014-06-11	Account
NULL	NULL	NULL	NULL	NULL	NULL

Result Grid Form Editor

11. Write an SQL query to fetch the count of employees working in the department ‘Admin’

```
SELECT COUNT(*) AS TOTAL_EMPLOYEES
FROM Worker
WHERE DEPARTMENT = 'Admin';
```



12. Write an SQL query to show the top n (say 10) records of a table

```
SELECT *
FROM Worker
LIMIT 10;
```